PANIMALAR ENGINEERING COLLEGE

(AN AUTONOMOUS INSTITUTION)

JAISAKTHI EDUCATIONAL TRUST
BANGALORE TRUNK ROAD, VARADHARAJAPURAM,
NASARATHPET, POONAMALLEE,
CHENNAI-600 123.

23ES1215- PROGRAMMING IN PYTHON LAB MANUAL

SEMESTER II

(2025-2026 - EVEN SEM)

PANIMALAR ENGINEERING COLLEGE

Jaisakthi Educational Trust
Bangalore Trunk Road, Varadharajapuram,
Nasarathpet, Poonamallee,
Chennai-600 123.

PROGRAM OUTCOMES

Engineering Graduates will be able to:

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.(WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5 and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

COURSE DETAILS

Course Code	Course Title	L	T	P	C
23ES1215	Programming In Python Lab	0	0	4	2

SYLLABUS

COURSE OBJECTIVE:

- To write, test, and debug simple Python programs
- To implement Python programs with conditions and loops
- To use functions for structuring Python programs.
- To represent compound data using Python lists, tuples, dictionaries.
- To learn to implement string functions and file operations
- To understand python packages and GUI development.

LIST OF EXPERIMENTS

1. Basic Python Programs
2. Write programs to demonstrate different number data types in python
3. Develop python programs to demonstrate various conditional statements
4. Implement user defined functions using python
5. Develop python scripts to demonstrate built-in functions
6. Develop python programs to perform various string operations like slicing, indexing &formatting
7. Develop python programs to perform operations on List & Tuple
8. Demonstrate the concept of Dictionary with python programs
9. Develop python programs to perform operations on Sets.

10. Develop python codes to perform matrix addition, subtraction and transpose of the given matrix
11. Develop python codes to demonstrate the concept of function composition and anonymous functions.
12. Demonstrate python codes to print try, except and finally block statements
13. Implement python programs to perform file operations
14. Write a python code to raise and handle various built in exceptions.
15. Implement python programs using packages numpy and pandas
16. UI development using tkinter

Mini Project :Suggested Topics(but not limited to)

1. Dice roll simulator
2. Guess the number game
3. Random password generator

TOTAL: 60 PERIODS

COURSE OUTCOME(S):

Upon successful completion of the course student will be able to:

CO1: Develop and execute simple Python programs

CO2: Implement programs in Python using conditionals and loops for solving problems.

CO3: Develop functions to decompose a Python program.

CO4: Compare various string operations in Python.

CO5: Experiment with Python packages in data analysis

CO6: Create GUI for python applications

WEB REFERENCES:

1. https://www.programiz.com/python-programming/examples
2. https://www.geeksforgeeks.org/python-programming-examples/
3. https://beginnersbook.com/2018/02/python-programs/
4. https://www.javatpoint.com/python-programs
5. https://www.w3schools.com/python/python_examples.asp

CO-PO MAPPING

CO's	COURSE OUTCOMES										
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO 1	3	3	3	2	2					2	
CO 2	3	3	3	2	2					2	
CO 3	3	3	3	3	3					3	
CO 4	3	3	3	3	3					3	
CO 5	3	3	3	3	3	2				3	
CO 6	3	3	3	3	3	3	3	2	2	3	2

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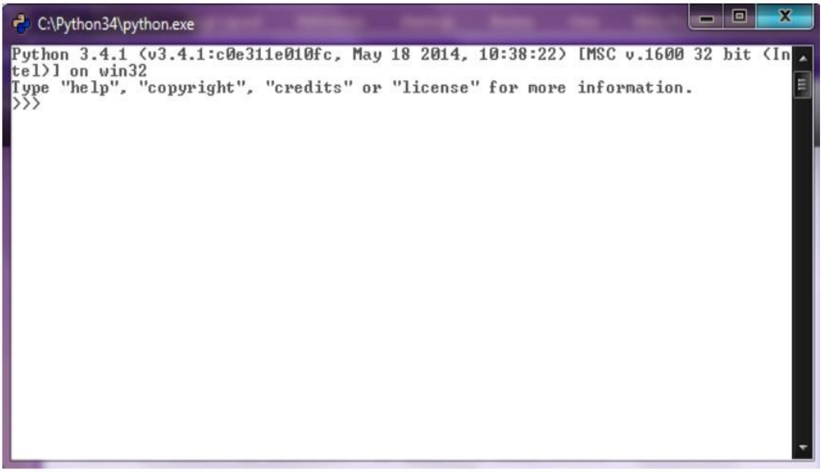
ADDITIONAL PROGRAMS

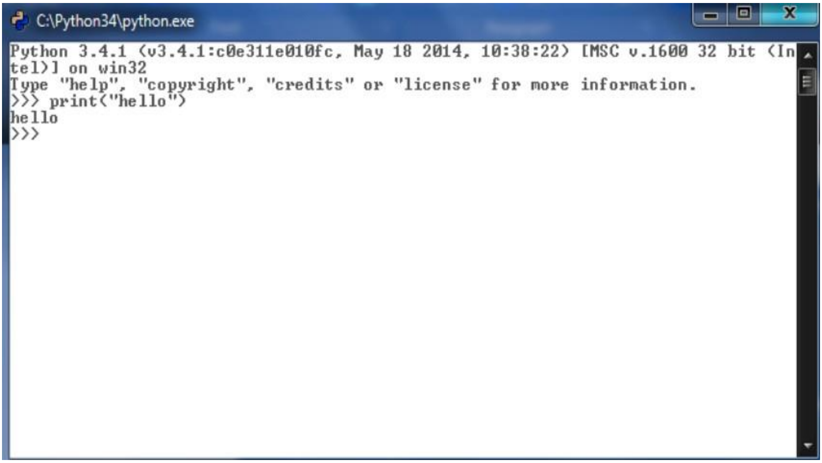
EXECUTION OF PYTHON PROGRAM USING WINDOWS

Python command line in Windows

STEP 1: Navigate to Start-> All Programs ->Python3.4->Python (Command Prompt)



STEP 2: Type the python code and press enter to execute the code.

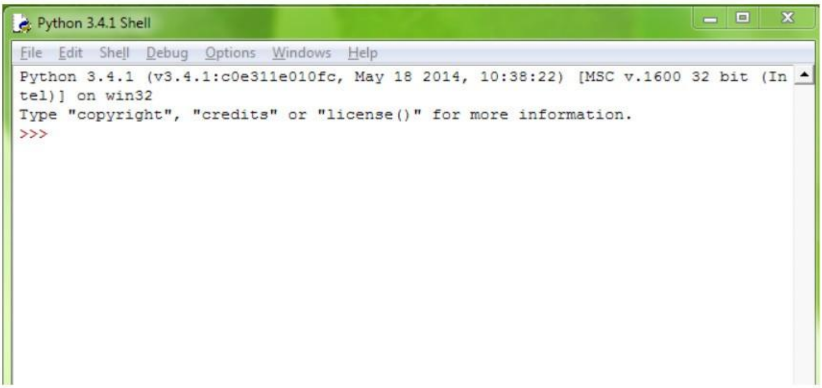


EXECUTION OF PYTHON PROGRAM USING GUI IN WINDOWS

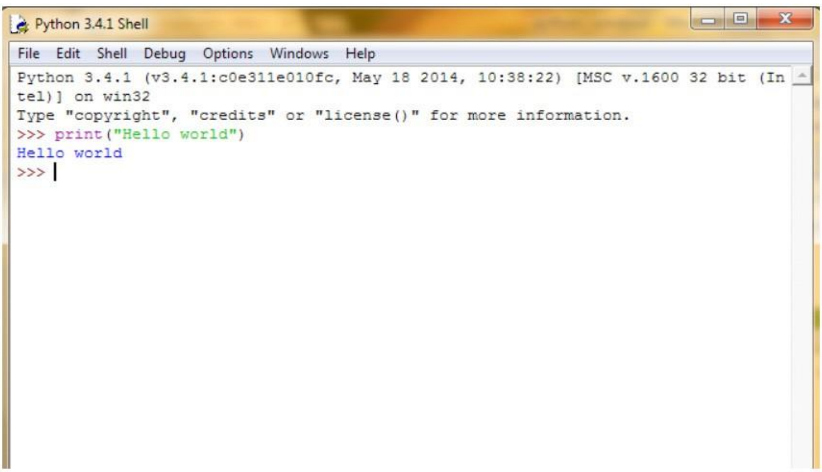
Python GUI in Windows

STEP 1: Navigate to Start-> All Programs ->Python3.4->IDLE(Python GUI).

The Python 3.4.1 Shell opens up.

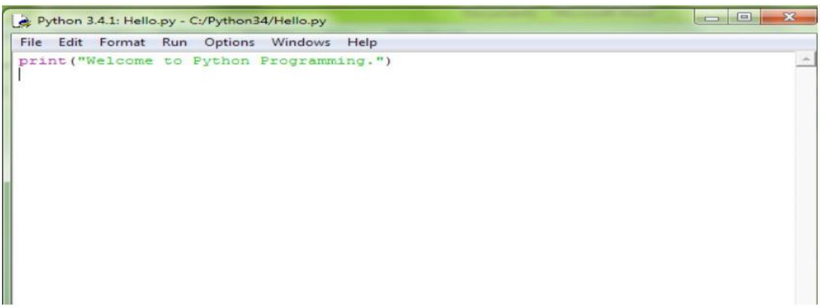


STEP 2: Type the python code in the shell and press enter to execute the code.

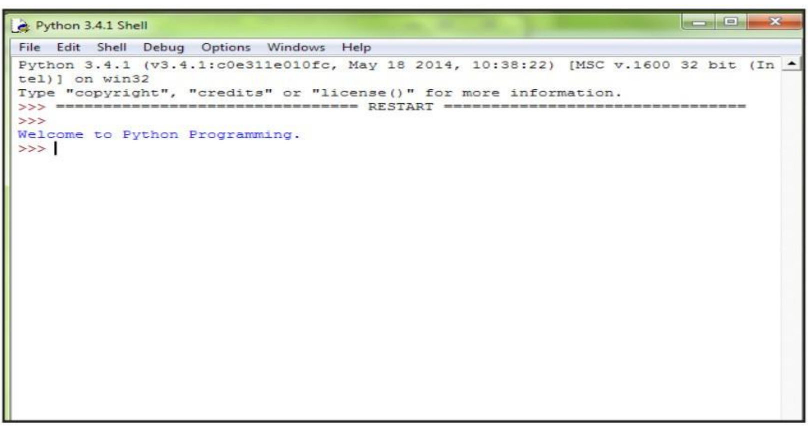


STEP 3: To save and execute code from a file; Navigate to File->New file to create a new file.

STEP 4: Type the program in the file. Navigate to File->Save As... and save the file. Python programs have py extension. Eg. Hello.py



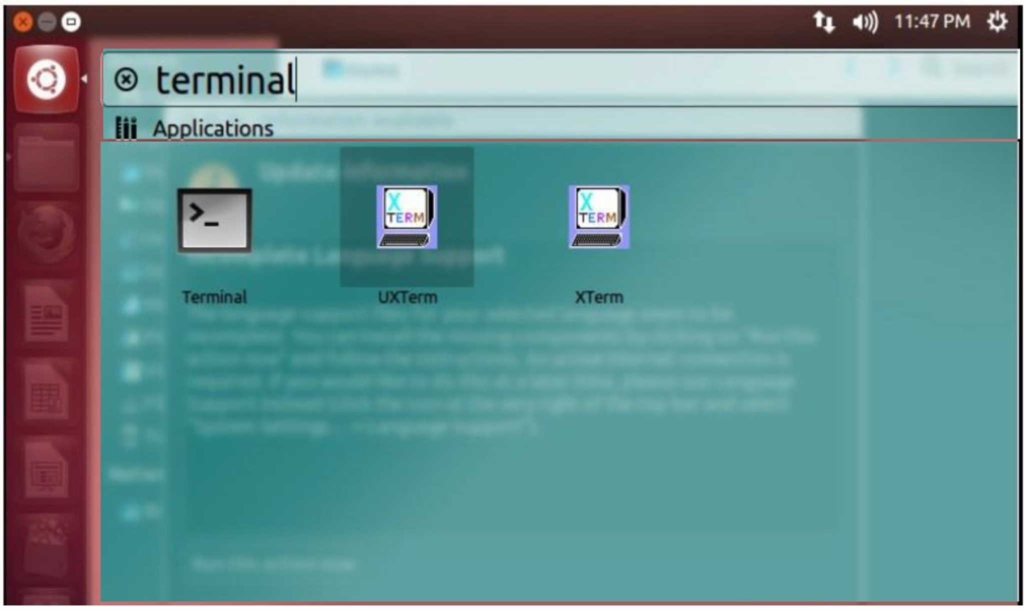
STEP 5: Press F5 or navigate to Run->Run Module, to execute the programs.



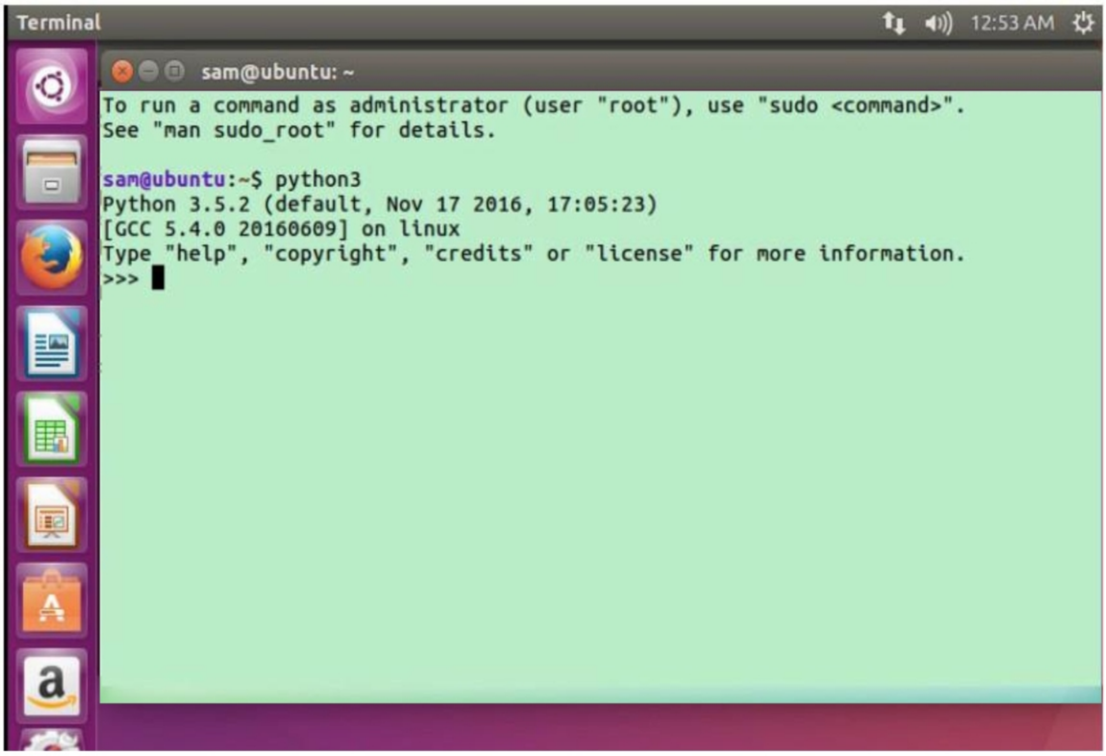
EXECUTION OF PYTHON PROGRAM USING UBUNTU

Python command line in Windows

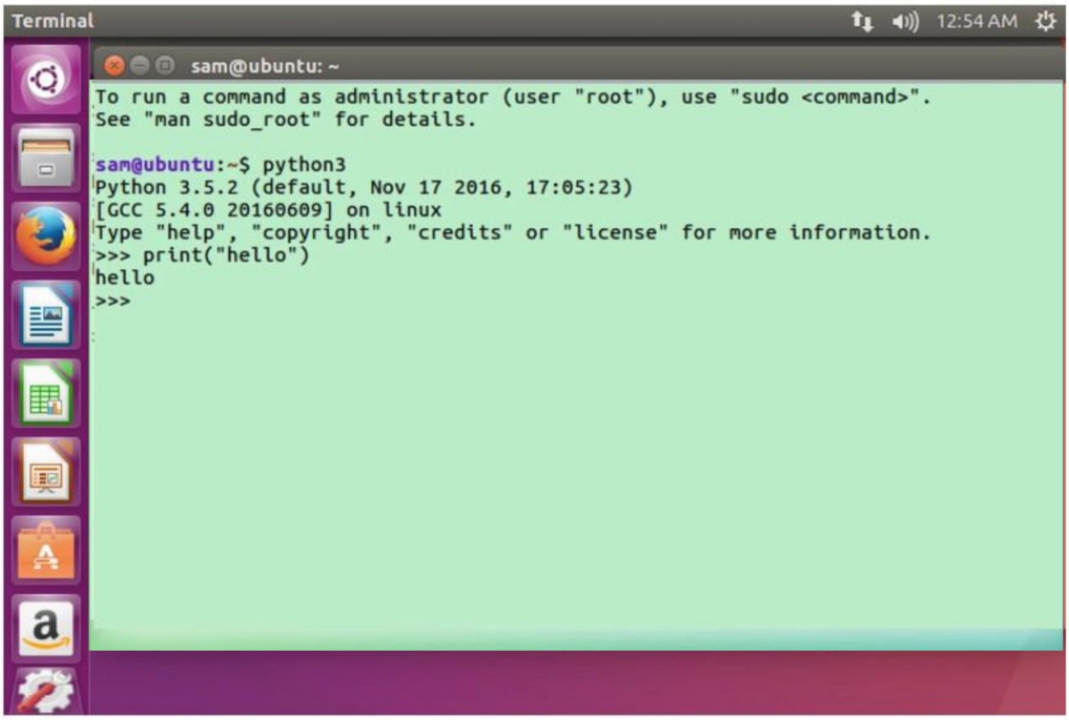
STEP 1: Click on "Dash Home" and type terminal in the search box to locate the "Terminal" application. Click on the "Terminal" application. The terminal opens up.



STEP 2: Type python3 to invoke the python 3 interpreter.

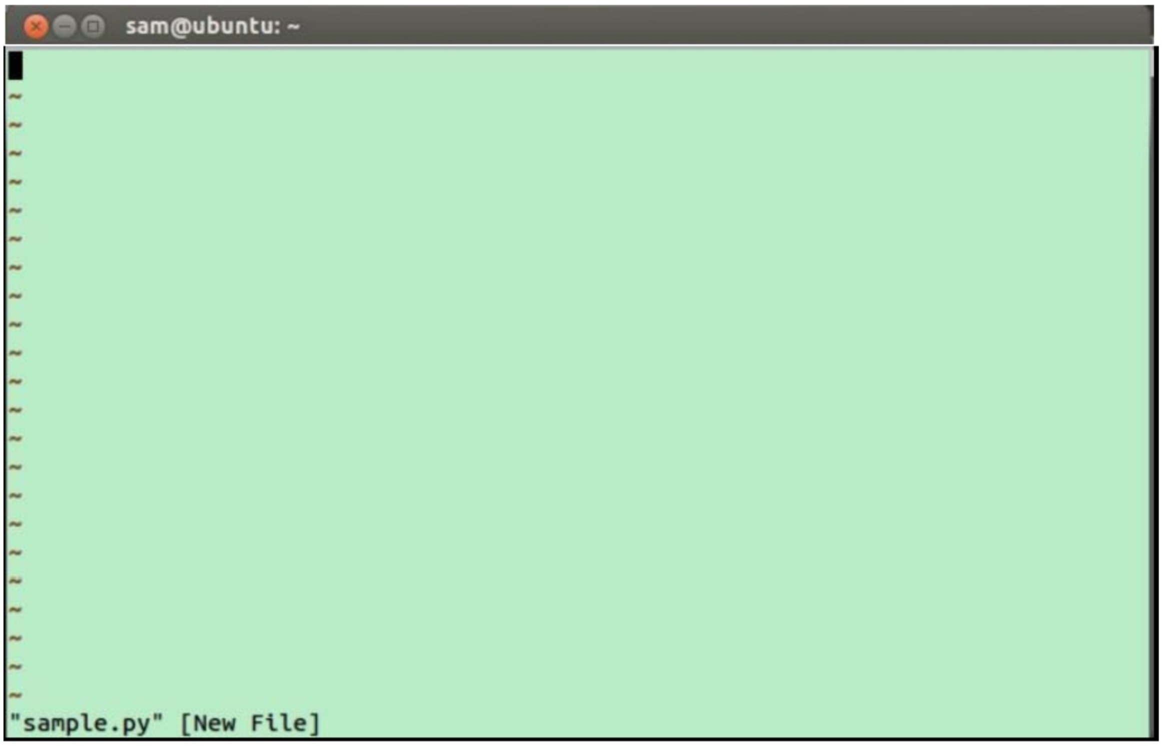


STEP 3: Type the python code and press enter to execute it.



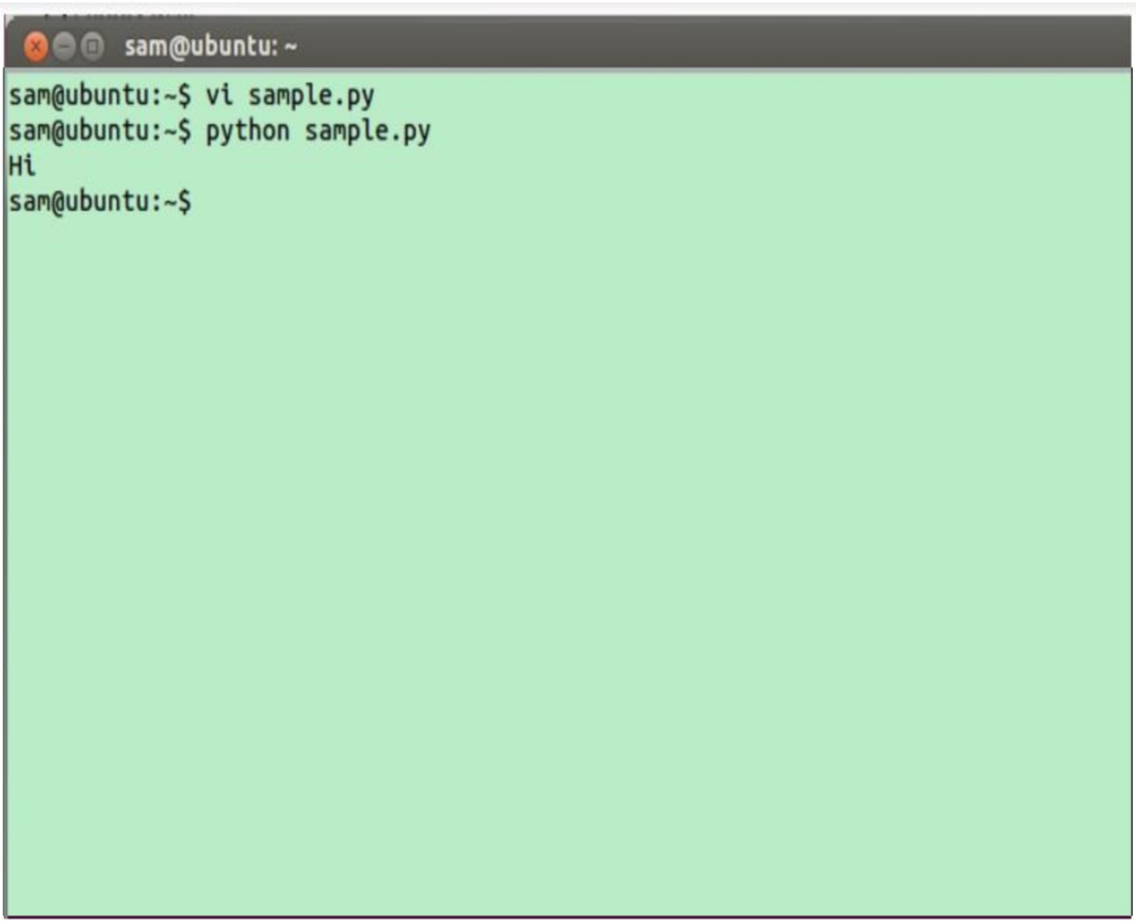
STEP 4: To create a python file, type vi filename.py

STEP 5: Type i to get into insert mode



STEP 6: Type the python code. Press esc key and type :wq to save and quit.

STEP 7: To execute the program, type python filename.py



EX NO. 1(A) TO COMPUTE DISTANCE TWO POINTS TAKING INPUT

AIM

Write a program to compute the distance between two points taking input from the user (Pythagorean Theorem).

ALGORITHM

Input: x1, y1, x2 and y2

Output: Distance between two points.

Step1: Start

Step2: Import math module

Step3: Read the values of x1, y1, x2, and y2

Step4: Calculate the distance using the formulae math.sqrt((x2 - x1)**2 + (y2 - y1)**2) and store the result in distance

Step5: Print distance

Step6: Stop

LOGIC OF THE PROGRAM

The Pythagorean theorem is the basis for computing distance between two points. Let (x1, y1) and (x2, y2) be the co-ordinates of points on xy-plane. From Pythagorean theorem, the distance between two points is calculated using the formulae:

$$d = \sqrt{(\Delta x)^2 + (\Delta y)^2} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

To find the distance, using the method of sqrt(). This method is not accessible directly, so to import math module and call this method using math static object. To find the power of a number, using ** operator.

Compute distance between two points:

- Read number of terms.
- Send values to mathematical sqrt function
- Print the distance

OUTPUT

Enter A Value: 7

Enter B Value: 8

Enter C Value: 11

Enter D Value: 15

The Distance is 8.06225774829855

RESULT

Thus, the python program to compute the distance between two points taking input from the user was written and executed successfully.

EX NO. 1(b) PROGRAM TO PERFORM ARITHMETIC OPERATIONS

AIM

Write a program to perform arithmetic operations like addition, subtraction, multiplication, division, modulus and exponent

ALGORITHM

Input: num1 & num2

Output: Add, Sub, Multiply, Divide, Modulus and Exponent of two points.

Step1: Start

Step2: Read the values of num1 and num2

Step3: Calculate the two points using arithmetic operations like Add, Sub, Multiply, Divide, Modulus and Exponent

Step4: Print the output

Step5: Stop

LOGIC OF THE PROGRAM

To write a computer program which can do simple calculation like adding two numbers (2 + 3) and to write a program, which can solve a complex equation like $P(x) = x^4 + 7x^3 - 5x + 9$.

examples: -

$2 + 3$

$P(x) = x^4 + 7x^3 - 5x + 9$.

These two statements are called arithmetic expressions in a programming language and plus, minus used in these expressions are called arithmetic operators and the values used in these expressions like 2, 3 and x, etc., are called operands.

OUTPUT

Please Enter the First Value Number 1: 17

Please Enter the Second Value Number 2: 20

The Sum of two numbers 37.0

The Subtraction of two numbers -3.0

The Multiplication of two numbers 340.0

The Division of two numbers 0.85

The Modulus of two numbers 17.0

The Exponent Value of two numbers 4.0642314066475725e+24

RESULT

Thus, the python program to perform arithmetic operations using two points taking input from the user was written and executed successfully.

EX NO. 2 PROGRAM TO DEMONSTRATE DIFFERENT NUMBER DATA TYPES IN PYTHON

AIM

Write a program to demonstrate different number data types in python.

ALGORITHM

Input: x

Output: To display different datatypes

Step1: Start

Step2: Read the values of num1 and num2

Step3: Print the output

Step4: Stop

LOGIC OF THE PROGRAM

Data type defines the kind of a value i.e the type of value whether it is int, string, float etc. Python enables us to check the type of the variable used in the program. Python provides us the type() function, which returns the type of the variable passed.

Numbers: Represent numeral values like int, float, complex

Int : It contains positive or negative whole numbers. eg: 1,2,3,-1.....

Float : Contains real floating point representation.
eg: 2.0,1.0,8.02....

Complex : Numbers of form Ai+Bj . eg: 2i+3j

Sequence Type:

String : A string is defined as a collection of one or more characters. It is putted in single quotes or double quotes or triple quotes i.e (‘ ’) , (“ ”) or (‘ ‘ ‘ ‘ ‘ ‘) . for eg: s=“GoEduHub.com”

List : It is as same as arrays, it contains heterogeneous datatypes. eg: a=[1,2,“a”]

Tuple : Tuples are created by ( ) . it contains heterogeneous datatypes . eg: t=(1,2,3)

Boolean: Booleans represent one of two values: True or False.

OUTPUT

20

<class ‘int’>

20.5

<class ‘float’>

1j

<class ‘complex’>

GoEduHub.com

<class ‘str’>

[‘Go’, ‘Edu’, ‘Hub’]

<class ‘list’>

(‘Python’, ‘from’, ‘GoEduHub.com’)

<class ‘tuple’>

True

<class ‘bool’>

b’Hello’

<class ‘bytes’>

RESULT

Thus, the python program to demonstrate different number datatypes was written and executed successfully.

EX NO. 3 DEVELOP A PYTHON PROGRAM TO DEMONSTRATE VARIOUS CONDITIONAL STATEMENTS

AIM

Write a program to develop a python program to demonstrate various conditional statements.

ALGORITHM

Input: a, lower, upper

Output: Display even number, factorial and list all prime numbers between interval

Step1: Start

Step2: Read the values of a, lower and upper

Step3: Using if condition to check the value is even or odd number

Step4: Print the output

Step5: Using for loop to display the factorial of given number

Step6: Print the output

Step7: Using for loop and if condition to display all the prime numbers between interval

Step8: Print the output

Step9: Stop

LOGIC OF THE PROGRAM

Checking whether the given number is even number or not

1. Read the number.
2. Divide the number by 2
3. Compare the reminder value
4. Display the result

Finding the factorial of a number

1. Read the number.
2. Decrement the value
3. Compare the reminder value
4. Display the result

Print the prime numbers between interval

1. Read the number.
2. Check the lower and upper number
3. Display all prime numbers between the intervals
4. End of the program.

OUTPUT

#Number is even number or not

12

Even

#Finding the factorial of a number

5

The Factorial Number is 120

#Print all Prime Numbers between an Interval

Enter lower range: 23

Enter upper range: 43

23

29

31

37

41

43

RESULT

Thus, the program python to demonstrate various conditional statements were written and executed successfully.

EX NO. 4 IMPLEMENT USER DEFINED FUNCTIONS USING PYTHON

AIM

Write a program to Implement user defined functions using python.

ALGORITHM

Input: a, b, sum

Output: sum of two numbers

Step1: Start

Step2: Read the user defined values of a, b

Step3: using arithmetic plus operation

Step4: Print the output

Step5: Stop

LOGIC OF THE PROGRAM

- A user-defined function's declaration begins with the keyword def and followed by the function name.
- The function may take arguments(s) as input within the opening and closing parentheses, just after the function name followed by a colon.
- After defining the function name and arguments(s) a block of program statement(s) start at the next line and these statement(s) must be indented.

OUTPUT

The Sum is: 7

RESULT

Thus, the python program to implement a user-defined function in the sum of two numbers was written and executed successfully.

EX NO. 5 DEVELOP A PYTHON SCRIPT TO DEMONSTRATE RANGE ( ) FUNCTION

AIM

Write a program to develop a python script to demonstrate range ( ) function.

ALGORITHM

Output: Display numbers and Sum of natural numbers

Step1: Start
Step2: Using for loop statement to display the number 0 to 9
Step3: Define the range of iteration using range()
Step4: Using for loop statement to display the sum of natural numbers

Step5: Print the output
Step6: Stop

LOGIC OF THE PROGRAM

range() allows user to generate a series of numbers within a given range. Depending on how many arguments user is passing to the function, user can decide where that series of numbers will begin and end as well as how big the difference will be between one number and the next.range() takes mainly three arguments.

start: integer starting from which the sequence of integers is to be returned

stop: integer before which the sequence of integers is to be returned. The range of integers end at stop – 1.

step: integer value which determines the increment between each integer in the sequence

OUTPUT

Printing a number

0 1 2 3 4 5 6 7 8 9

Using range for iteration

10 20 30 40

Performing sum of natural number

Sum of first 10 natural number: 55

RESULT

Thus, the python program to implement a range() function in the sum of natural numbers was written and executed successfully.

EX NO. 6 PROGRAM TO PERFORM VARIOUS STRING OPERATIONS LIKE SLICING, INDEXING & FORMATTING

AIM

Write a program to develop a python program to perform various string operations like slicing, indexing & formatting.

ALGORITHM

Output: Slicing, indexing, and formatting string value

Step1: Start

Step2: Starting index where the slicing of an object starts

Step3: Ending index where the slicing of object stops

Step4: It is an optional argument that determines the increment between each index for slicing.

Step5: Indexing syntax can be used as a substitute for the slice object.

Step6: Print the output

Step7: Stop

LOGIC OF THE PROGRAM

String slicing in Python is to obtain a substring from the main string. Slice of string means part (substring) of string.

<String Name> [Start: Stop: Step]

Start: A position to start the cutting of the string.

Stop: A position before to cut the string.

Indexing

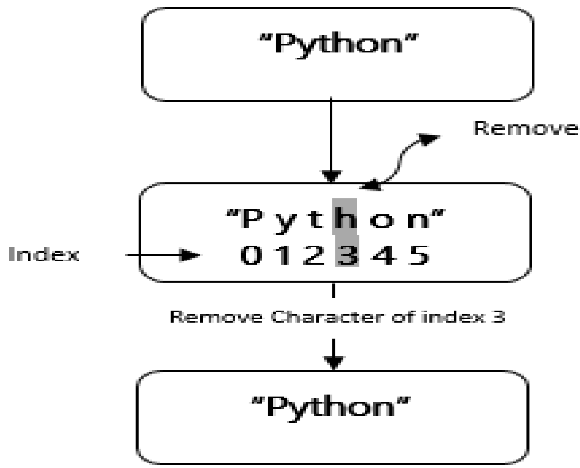
Program to remove the n*th* index character from a non-empty string.

Formatting

The format() method formats the specified value(s) and insert them inside the string's placeholder.

The placeholder is defined using curly brackets: {}. Read more about the placeholders in the Placeholder section below.

The format() method returns the formatted string.



OUTPUT

Output of String Slicing:

imalar Engineer

nmlrEg

Panimala

r Engineering Colleg

Pil

 niern olg

o

rnnrgo

Output of String Indexing:

ython

Pyton

Pytho

Output of String Formatting:

My name is John, I’m 36

My name is John, I’m 36

My name is John, I’m 36

RESULT

Thus, the python program to implement string slicing, indexing, and formatting was written and executed successfully.

EX NO. 7 DEVELOP PYTHON PROGRAM TO PERFORM OPERATIONS ON LIST & TUPLE

AIM

Write a program to develop python program to perform operations on List & Tuple.

ALGORITHM

Input: List, Tuple

Output: Display length, min, max, sum, and sort of list and tuple value

Step1: Start

Step2: Read the value of List and Tuple

Step3: Use buitin functions to find length, min, max, sum, and sort of list and tuple

Step4: Print the output

Step5: Stop

LOGIC OF THE PROGRAM

List and Tuple are built-in container types defined in Python. Objects of both these types can store different other objects that are accessible by index. List as well as tuple are sequence data type, just as string. List as well as tuple can store objects which need not be of same type.

List: A List is an ordered collection of items (which may be of same or different types) separated by comma and enclosed in square brackets.

Tuple: Tuple looks similar to list. The only difference is that comma separated items of same or different type are enclosed in parentheses. Individual items follow zero based index, as in list or string.

OUTPUT

Length of the List: 5

Length of the Tuple: 5

max of List 50

max of Tuple 50

min of List 10

min of Tuple 10

sum of List 150

sum of Tuple 150

List in sorted order [10, 20, 30, 40, 50]

Tuple in sorted order [10, 20, 30, 40, 50]

RESULT

Thus, the python program to perform various methods of the list and the tuple value was written and executed successfully.

EX NO. 8 DEMONSTRATE THE CONCEPT OF DICTIONARY WITH PYTHON PROGRAM

AIM

Write a program to demonstrate the concept of a dictionary with a python program.

ALGORITHM

Output: Integer key and values

Step1: Start

Step2: Read the value ‘a’

Step3: pass the *dictionary name* in the *print* function

Step4: To print the dictionary *keys* only, loop through the dictionary and print it

5: To print the dictionary values only, loop through the dictionary and use the *values()* method, and it returns the dictionary values.

Step6: Print the output

Step7: Stop

LOGIC OF THE PROGRAM

A dictionary contains the pair of the keys & values (represents *key: value*), a dictionary is created by providing the elements within the curly braces ({}), separated by the commas.

Here, dictionary is created with integer keys and printing the dictionary, it’s keys, it’s values and key-value pairs.

Printing the dictionary – To print a dictionary, just pass the *dictionary_name* in the *print* function.

Syntax:

```
print(dictionary_name)
```

Printing the keys – To print the dictionary *keys* only, loop through the dictionary and print it, while looping through a dictionary it returns the keys.

Syntax:

```
for key in dictionary_name:
    print(key)
```

Printing the values – To print the dictionary values only, loop through the dictionary and use the *values()* method, it returns the dictionary values. Syntax:

```
for value in dictionary_name.values():
    print(value)
```

Print the *key-value* pairs – To print the keys & values both, loop through the dictionary and use the *items()* method, it returns the key-value pair.

Syntax:

```
for key,value in dictionary_name.items():
    print(key, value)
```

OUTPUT

Dictionary 'dict_a' is...

{1: 'India', 2: 'USA', 3: 'UK', 4: 'Canada'}

Dictionary 'dict_a' keys...

1

2

3

4

Dictionary 'dict_a' values...

India

USA

UK

Canada

Dictionary ‘dict_a’ keys & values...

1 : India

2 : USA

3 : UK

4 : Canada

RESULT

Thus, the python program to implement a dictionary with integer key and value was written and executed successfully.

EX NO. 9 DEVELOP PYTHON PROGRAMS TO PERFORM OPERATIONS ON SETS

AIM

Write a program to perform operations on Sets

ALGORITHM

Input: Integer values

Output: Set operations

Step1: Start

Step2: Define the values of the sets

Step3: Using the set operations like union, intersection, difference, and symmetric difference.

Step4: Print the output

Step5: Stop

LOGIC OF THE PROGRAM

Set:

A set is a changeable object that contains an unordered collection of things with only unique values. The set structure is valuable because it includes a variety of mathematical operations that may be used to work with the data stored in a set. A set is formed by enclosing all of the items (elements) behind curly brackets {}, separated by commas, or by using the built-in set() function.

Operations:

Sets can be used to perform set operations such as union, intersection, difference, and symmetric difference.

Union Operation:

Union set operation performs with 2 functions i.e. it is the set of all the elements from both the sets in a single different set. The | operator is used to perform union. The union() method can be used to do the same thing. If similar elements occur in both the sets, then their single value is considered.

Intersection Operation:

The intersection of two sets is a set of elements that are shared by both sets i.e. generate a list of elements that are common between two sets. The & operator is used to perform intersection. The intersection() method can be used to do the same thing.

Difference Operation:

The difference between sets B and A (i.e. A-B) is a set of elements that are only in A but not in B. The – operator is used to perform a difference. The difference() method can be used to do the same thing. The difference() method can be used to determine which elements are present in one set but not in another.

Symmetric Difference Operation:

A and B Symmetric Difference is a set of items in A and B but not in both (excluding the intersection). The operator is used to perform symmetric difference. The same thing can be done with the symmetric_difference() method.

OUTPUT

Union of A and B: {0, 2, 4, 6, 7, 9, 15, 17, 19, 22, 23, 31}
Difference of A and B: {15, 17, 19, 22, 23, 31}
Intersection of A and B: {7}
Symmetric Difference of A and B: {0, 2, 4, 6, 9, 15, 17, 19, 22, 23, 31}

RESULT

Thus, the python program to perform various operations on sets was written and executed successfully.

EX NO. 10 DEVELOP FOR PYTHON CODES TO PERFORM MATRIX ADDITION AND SUBTRACTION AND TRANSPOSE

AIM

Write a program to perform matrix addition, subtraction and transpose.

MATRIX ADDITION AND SUBTRACTION

ALGORITHM

Input: Two input matrix

Output: Addition of two matrix

Step1: Define two matrices matrix1 and matrix2.

Step 2: Print the elements of matrix1 and matrix2 using nested loops.

Step 3: Define an empty matrix result of the same size as matrix1.

Step 4: Add, Subtract the corresponding elements of matrix1 and matrix2 using nested loops and store the result in the result matrix.

Step5: Print the result matrix.

OUTPUT

Printing elements of first matrix

1 2

3 4

Printing elements of second matrix

4 5

6 7

Subtraction of two matrix

-3 -3

-3 -3

Addition of two matrix

5 7

9 11

MATRIX TRANSPOSE

ALGORITHM:

Input: Input matrix

Output: Transpose of matrix

Step 1:Define two matrices matrix1

Step 2:Print the elements of matrix1 using nested loops.

Step 3:Define an empty matrix result of the same size as matrix1.

Step 4:Transpose the corresponding elements of matrix1 using nested loops and store the result in the result matrix.

Step 5:Print the result matrix.

OUTPUT

Printing elements of first matrix

1 2

3 4

Subtraction of two matrix

1 3

2 4

RESULT

Thus, the python program to perform matrix addition , subtraction and transpose was written and executed successfully.

EX NO. 11 DEVELOP PYTHON CODES TO DEMONSTRATE THE CONCEPT OF FUNCTION COMPOSITION AND ANONYMOUS FUNCTIONS

AIM:

Write a program to demonstrate the concept of function composition and anonymous functions.

ALGORITHM:

Input: Two input functions

Output: Output of combine two input functions

Step1: Start

Step 2: Define the function name like as composite_function() with argumetns

Step 3: Define two more function as add and ,multiply

Step 4: combine thesse two function using function compostion

Step5: Print the result

LOGIC OF THE PROGRAM

FUNCTION COMPOSITION

Function composition is the way of combining two or more functions in such a way that the output of one function becomes the input of the second function and so on. For example, let there be two functions "F" and "G" and their composition can be represented as F(G(x)) where "x" is the argument and output of G(x) function will become the input of F() function.

ANONYMOUS FUNCTIONS(Lambda Function)

Syntax:

lambda arguments : expression

This function can have any number of arguments but only one expression, which is evaluated and return

OUTPUT

result : 14

RESULT

Thus, the python program to demonstrate the concept of function composition and anonymous functions was written and executed successfully.

EX NO. 12 DEMONSTRATE A PYTHON CODE TO PRINT TRY, EXCEPT AND FINALLY BLOCK STATEMENTS

AIM

Write a program to demonstrate a python code to print try, except and finally block statements.

ALGORITHM

Input: num1, num2, result
Output: sum of two values
Step1: Start
Step2: Read the input values
Step3: using try statement to run the function sum of two a given values
Step4: Exceptions occurs, print the error message in the catch statement
Step5: finally statement to print the execution message
Step6: Stop

LOGIC OF THE PROGRAM

- Exception handling enables to handle errors gracefully and do something meaningful about it. Like display a message to user if intended file not found. Python handles exception using try, except block.
- First statement between try and except block are executed.
- If no exception occurs then code under except clause will be skipped.
- try:If file don't exist then exception will be raised and the rest of the code in the try block will be skipped
- When exceptions occur, if the exception type matches exception name after except keyword, then the code in that except clause is executed.
- A try statement can have more than once except clause or it can also have an optional else and/or finally statement.

```
  <body>
    except <ExceptionType1>:
        <handler1>
    except <ExceptionTypeN>:
        <handlerN>
    except:
        <handlerExcept>
    else:
        <process_else>
finally:
        <process_finally>
```

OUTPUT

Enter two numbers, separated by a comma: 5,9

Result is 0.5555555555555556

No exceptions

This will execute no matter what

RESULT

Thus, the python program to perform exception handling to check whether the program was written and executed successfully.

EX NO. 13 IMPLEMENT PYTHON PROGRAM TO PERFORM FILE OPERATIONS

AIM

Write a program to implement python program to perform file operations.

ALGORITHM

Output: File create, read, write and append

Step1: Start

Step2: Create a new text file “file1.txt” using the mode of “x”

Step3: Write some text in the new text file “file1.txt” using write mode “w”

Step4: Read a message from the existing text file “file1.txt” using read mode “r”

Step5: Append a message to the existing text file “file1.txt” using append mode “a”

Step6: Stop

LOGIC OF THE PROGRAM

Files are named locations on disk to store related information. They are used to permanently store data in a non-volatile memory (e.g., hard disk).

Since Random Access Memory (RAM) is volatile (which loses its data when the computer is turned off), files are used for future purpose for string data permanently storing them. To read or write to a file, first open the file. Once opened and completed, it needs to be closed so that the resources that are tied with the file are freed.

Hence, in Python, a file operation takes place in the following order:

- Open a file
- Read or write (perform operation)
- Close the file

Opening Files in Python

Python has a built-in open() function to open a file. This function returns a file object, also called a handle, as it is used to read or modify the file accordingly. File mode operation can be specified as; for read ‘r’, for write ‘w’, for append ‘a’, respectively. File can also open in text mode or binary mode.

OUTPUT

Panimalar Engineering College

Readline:

Panimalar

Successfully Inserted

RESULT

Thus, the python program using file operation to create, read, write and append messages was written and executed successfully.

EX NO. 14 DEVELOP PYTHON CODES RAISE AND HANDLE VARIOUS BUILT IN EXCEPTIONS

AIM

Write a program to demonstrate the concept various built in exception handling functions

ALGORITHM

Output: Raise and handle various Built in exceptions

Step1: Start

Step 2: try block tests the statement of error.

Step 3: this try block contains multiple statements that may throw different types of exceptions.

Step 4: if there is any error in that try block, it raises the error in that block.

Step5: except block handle the error using exception handling built-in functions that error occurs in try block.

Step 6: Stop

LOGIC OF THE PROGRAM

Exception Value Error:

Raised when an operation or function receives an argument that has the right type but an inappropriate value, and the situation is not described by a more precise exception such as Index Error

exception FileNotFoundError:

Raised when a file or directory is requested but doesn't exist.

exception TypeError:

Raised when an operation or function is applied to an object of inappropriate type. The associated value is a string giving details about the type mismatch.

exception IndexError:

Raised when a sequence subscript is out of range. Slice indices are silently truncated to fall in the allowed range; if an index is not an integer, TypeError is raised.

OUTPUT

Input an integer: name

Error: Invalid input, input a valid integer. Input value: None

Input an integer: 5.56

Error: Invalid input, input a valid integer. Input value: None

Input an integer: 51

Input value: 51

a) Program to handles and raise a FileNotFound Error exception if the file does not exist.

OUTPUT

Input a files name: test.txt

Error: File not found.

Input a file name: file2.txt

File contents

b) Program to handles and raise a TypeError exception if the inputs are not numerical.

OUTPUT

Input the first number: a

Error: Invalid input. Please Input a valid number.

Input the first number:12

Input the second number: b

Error: Invalid input. Please Input a valid number.

Input the second number: 12

Product of the said two numbers: 24.0

c) Program to handles an IndexError exception if the index is out of range.

OUTPUT

Input the index: 11

Error: Index out of range.

Input the index: 0

Result: 1

Input the index: 4

Result: 5

RESULT

Thus, the python program to demonstrate the concept of various built in exception handling functions was written and executed successfully.

EX NO. 15 DEVELOP PYTHON PROGRAMS USING PACKAGES NUMPY AND PANDAS

AIM

Write programs using packages numpy and pandas.

ALGORITHM

Input: Array of Matrix

Output: Data frames of pandas

Step1: Start

Step 2: to create the package of numpy and array index in numpy for numerical calculation.

Step 3: to create the package of pandas and call dataframe function in pandas.

Step 4: Print the result

Step5: Stop

LOGIC OF THE PROGRAM

In command prompt install this packages

1. pip install numpy

2, pip install pandas

OUTPUT

```
   x  y z
A  5 22 3
B 13 24 6
```

RESULT

Thus, the python programs using packages numpy and pandas was written and executed successfully.

EX NO. 16 DEVELOP PYTHON PROGRAMS USING TKINTER

AIM

Write a program to develop GUI application using tkinter

ALGORITHM

Input: Various input tools of tkinter

Output: Output of various application

Step1: Start

Step 2: develop the application using tkinter widgets

Step 3: widgets like button ,label, entry are used to develop the applications

Step 4: Run the application

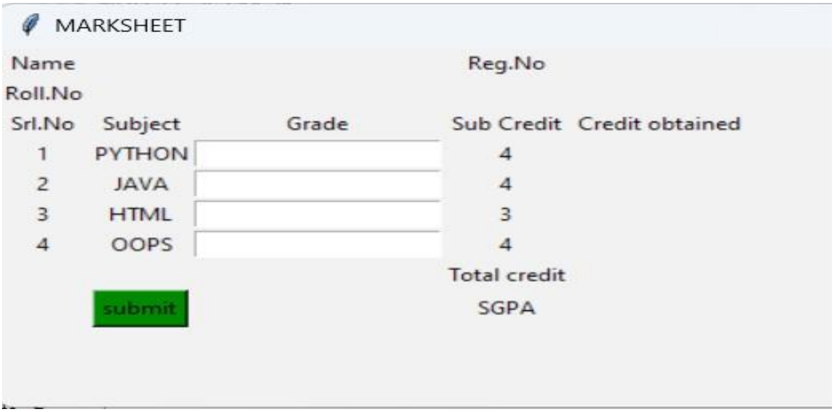
Step 5:Stop

LOGIC OF THE PROGRAM

In command prompt install this packages

1. pip install tkinter

OUTPUT



RESULT

Thus, the python program to develop GUI application using tkinter was written and executed successfully.

ADDITIONAL PROGRAMS

EX NO. A1 PROGRAM TO FIND THE BIGGEST OF THREE NUMBERS

AIM

To write a program in python to find the biggest of three numbers.

ALGORITHM

Step 1: Enter the three numbers

Step 2: use the following if condition to find the biggest number

Step 3: if (num1 > num2) and (num1 > num3): Number 1 is the largest

Step 4: elif (num2 > num1) and (num2 > num3): Number 2 is largest

Step 5: else number 3 is largest

Step 6: Print the output

LOGIC OF THE PROGRAM

Program is to enter the three numbers as input and use if else condition to find the largest number

Ex. if (num1 > num2) and (num1 > num3):

Number 1 is the largest

elif (num2 > num1) and (num2 > num3):

Number 2 is largest

else number 3 is largest

OUTPUT

The largest number between 10, 14 and 12 is 14

RESULT

Thus, the program in python to find the biggest of three numbers was written and executed successfully.

EX NO. A2 PROGRAM TO CHECK THE GIVEN NUMBER IS EVEN OR ODD

AIM

To write a program in python to check the given number is even or odd.

ALGORITHM

Step 1: Read the number

Step 2: Then divide the number by 2

Step 3: if there is a remainder then the number is odd else the number is even

Step 4: Print the output

LOGIC OF THE PROGRAM

Enter the number

Check if the number is divided by 2

If there is a remainder then the number is odd or the number is even

OUTPUT

Enter number:124

Number is even!

Enter number:567

Number is odd!

RESULT

Thus, the program in python to check the given number is odd or even was written and executed successfully.

EX NO. A3 PROGRAM TO FIND FACTORIAL OF A NUMBER USING WHILE LOOP

AIM

To write a program in python to find factorial of a number using while loop.

ALGORITHM

Step 1: Get the input from the user

Step 2: Assume fac = 1

Step 3: And i=1

Step 4: Initialize the while loop

Step 5: Print the output

LOGIC OF THE PROGRAM

The factorial of a number is the product of all the integers from 1 to that number. For example, the factorial of 6 (denoted as 6!) is 1*2*3*4*5*6 = 720. Factorial is not defined for negative numbers and the factorial of zero is one, 0! = 1.

OUTPUT

enter a number: 4 factorial of 4 is 24

RESULT

Thus, the program in python to find the factorial of the number using while loop was written and executed successfully.

EX NO. A4 PROGRAM TO FIND FACTORIAL OF A NUMBER USING FOR LOOP

AIM

To write a program in python to the factorial of the number using for loop.

ALGORITHM

Step 1: Get the input from the user

Step 2: Assume factorial = 1

Step 3: If the number is negative print factorial is not available

Step 4: In else condition use the condition for factorial

Step 5: print the output

LOGIC OF THE PROGRAM

The factorial of a number is the product of all the integers from 1 to that number. For example, the factorial of 6 (denoted as 6!) is 1*2*3*4*5*6 = 720. Factorial is not defined for negative numbers and the factorial of zero is one, 0! = 1.

OUTPUT

The factorial of 5 is 120

RESULT

Thus, the program in python to find the factorial of the number using for loop was written and executed successfully.

EX NO. A5 PROGRAM TO PRINT REVERSE OF A NUMBER

AIM

To write a program in python to print the reverse of a number.

ALGORITHM

Step 1: Get the input from the user

Step 2: Assume reverse as 0

Step 3: Then initialize the while condition

Step 4: Assume Reminder = num %10

Reverse = (Reverse *10) + Reminder

num = num //10

Step 5: Print the output

LOGIC OF THE PROGRAM

Get the number

while(num > 0):

Reminder = num %10

Reverse = (Reverse *10) + Reminder

num = num //10

then finally print the number

OUTPUT

Enter any number: 231

Reverse of entered number is: 132

RESULT

Thus, the program in python to find the reverse of the number was written and executed successfully.

EX NO. A6 PROGRAM TO CHECK THE GIVEN STRINGS IS PALINDROME OR NOT

AIM

To write a program in python to check the given strings is palindrome or not.

ALGORITHM

Step 1: Get the input from the user

Step 2: Assume

index=0 check=True

Step 3: Initialize the while condition

Step 4: Initialize the if statement

```
if n[index]==n[-1-index]:
 index+=1
return True
return False
```

Step 5: Print the output

LOGIC OF THE PROGRAM

A **palindrome** is a word, phrase, number, or other sequence of characters which reads the same backward as forward, suchas madam or racecar.

```
while index<len(n):
if n[index]==n[-1-index]:
index+=1
return True
return False
```

OUTPUT

Enter the string: Madam
It is a palindrome

RESULT

Thus, the program in python to check the palindrome condition was written and executed successfully.

EX NO. A7 PROGRAM TO CHECK GIVEN NUMBER IS PRIME OR NOT

AIM

To write a program in python to check given number is prime or not

ALGORITHM

Step 1: Get the number from the user and the number should be greater than 1

Step 2: Then check for the factors using for loop

Step 3: if (num % i) == 0:
The number is not a prime number

Step 4: Else the number is a prime number

Step 5: Print the output

LOGIC OF THE PROGRAM

A Prime Number can be divided evenly only by 1, or itself. And it must be a whole number greater than 1.

Example: 5 can only be divided evenly by 1 or 5, so it is a prime number.

But 6 can be divided evenly by 1, 2, 3 and 6 so it is NOT a prime number (it is a composite number).

OUTPUT

407 is not a prime number
11 times 37 is 407
103 is a prime number.

RESULT

Thus, the program in python to check the given number is prime or not was written and executed successfully.

EX NO. A8 PROGRAM TO GENERATE FIBONACCI SERIES

AIM

To write a program in python to generate the Fibonacci series.

ALGORITHM

Step 1: Get the n as input from the user

Step 2: Then assume n1 = 0
n2 = 1
i = 2

Step 3: Initialize the while loop as per the condition

Step 4: Then update the valuesn1=n2
n2=n3
i=i+1

Step 5: Finally print the output

LOGIC OF THE PROGRAM

Fibonacci Series or Fibonacci Numbers are the numbers that are displayed in following sequence

Fibonacci Series = 0, 1, 1, 2, 3, 5, 8, 13, 21, 34 …

If you observe the above pattern, First Value is 0, Second Value is 1 and the subsequent number is the result of sum of the previous two numbers. For example, third value is (0 + 1), Fourth value is (1 + 1) so on and so forth

OUTPUT

Fibonacci sequence upto 10 :
0, 1, 1, 2, 3, 5, 8, 13, 21, 34,

RESULT

Thus, the program in python to generate Fibonacci series was written and executed successfully.

EX NO. A9 PROGRAM FOR FLOYDS TRIANGLE

AIM

To write a program in python to the Floyds triangle.

ALGORITHM

Step 1: Initialize count as 0

Step 2: Assume the range as 1to 6 using for loop

Step 3: Initialize the j loop

Step 4: Now assume count = count +1

Step 5: Finally print the count

LOGIC OF THE PROGRAM

Floyd’s triangle lists the natural numbers in a right triangle aligned to the left where the first row is **1** (unity) successive rows start towards the left with the next number followed by successive naturals listing one more number than theline above.

The first few lines of a Floyd triangle looks like this:

```
1

     2    3
     4    5    6
     7    8    9    10
11   12   13   14   15
```

OUTPUT

```
1

2    3

4    5    6

7    8    9    10
```

RESULT

Thus, the program in python to find the Floyds triangle was written and executed successfully.

EX NO. A10 PROGRAM FOR BUBBLE SORT

AIM

To write a program in python to implement the bubble sort.

ALGORITHM

Step 1: Define the function for bubble sort

Step 2: Usethe forloopas forpassnumin range(len(nlist)-1,0,-1):to enter the numbers

Step 3: Then within this use the if statement as if nlist[i]>nlist[i+1]:

Step 4: Compute temp = nlist[i] nlist[i] = nlist[i+1] nlist[i+1] = temp

Step 5: Print the output

LOGIC OF THE PROGRAM

Bubble sort, sometimes referred to as sinking sort, is a simple sorting algorithm that repeatedly steps through the list to be sorted, compares each pair of adjacent items and swaps them if they are in the wrong order. The pass through the list is repeated until no swaps are needed, which indicates that the list is sorted.

OUTPUT

14 , 21, 27, 41, 43, 45, 46, 57, 70

RESULT

Thus, the program in python to implement the bubble sort was written and executed successfully.

EX NO. A11 PROGRAM TO FIND SUM OF 'N' NATURAL NUMBERS

AIM

To write a program in python to find the sum of n natural numbers.

ALGORITHM

Step 1: Get the number up to you need to find from the user

Step 2: Check whether the number is positive or print sum =0

Step 3: Compute the while loop

Step 4: sum += num
num -= 1

Step 5: Calculate and print the output

LOGIC OF THE PROGRAM

Here, we store the number in num and display the sum of natural numbers up to that number. We use while loop to iterate until the number becomes zero.

OUTPUT

The sum is 171

RESULT

Thus, the program in python to find the sum of n natural numbers was written and executed successfully.

EX NO. A12 PROGRAM USING BREAK & CONTINUE STATEMENT

AIM

To write a program in python to implement the program using break & continue statement.

ALGORITHM

Step 1: Initialize the number as 0

Step 2: Then within the for loop give the if condition as if number == 5: break here

Step 3: For the continue statement, Initialize the number as 0

Step 4: Provide the continue statement as per the syntaxPrint out of loop

Step 5: Finally print the output

LOGIC OF THE PROGRAM

BREAK STATEMENT

In Python, the break statement provides an opportunity to exit out of a loop when an external condition is triggered.

CONTINUE STATEMENT

The continue statement gives the option to skip over the part of a loop where an external condition is triggered, but to go on to complete the rest of the loop. That is, the current iteration of the loop will be disrupted, but the program will return to the top of the loop.

The continue statement will be within the block of code under the loop statement, usually after a conditional if statement.

OUTPUT

 # Break Statement

Number is 1

Number is 2

Number is 3

Number is 4

Out of loop

 # Continue Statement

Number is 1

Number is 2

Number is 3

Number is 4

Number is 6

Number is 7

Number is 8

Number is 9

Number is 10

Out of loop

RESULT

Thus, the program in python to implement the program using break & continue statement was written and executed successfully.

EX NO. A13 PROGRAM TO FIND LEAP YEAR OR NOT

AIM

To write a program in python to find the leap year or not a leap year.

ALGORITHM

Step 1: Read the input from the user

Step 2: Then use the if condition of if (year % 4) = = 0:Then if (year % 100) = = 0: Then if (year % 400) = = 0:

Step 3: If the above condition is satisfied then the year is leap year

Step 4: else the year is not a leap except for the particular century year

Step 5: Check the condition for the given year

Step 6: Print the output

LOGIC OF THE PROGRAM

If the year is divisible by 4 then it is leap year except the century years (year that have 00 in end). A century year is leap year only if it is not divisible by 100 but divisible by 400.

For example, 2000, 2016 are leap year while 2002, 2017 are not leap year.

To find the year is leap year or not check whether the year % 4= =0 And year %100 = = 0 Year % 400 = = 0 Use of **format** function: This lets you concatenate elements together within a string through positional formatting.

OUTPUT

2020 is a leap year

RESULT

Thus, the program in python to find the leap year was written and executed successfully.

EX NO. A14 PROGRAM TO FIND THE AREA AND CIRCUMFERENCE OF A CIRCLE

AIM

To write a program in python to find the area and circumference of a circle.

ALGORITHM

Step 1: Define the value of pi

Step 2: Read the value of pi from the user

Step 3: Enter the formula for the area and the circumference of the circle

Step 4: Calculate the area and circumference

Step 5: Print the value of the area and circumference of the circle

LOGIC OF THE PROGRAM

The formula for the radius of the circle is pi * r*r

The formula for the circumference of the circle is 2* pi*r

OUTPUT

Enter the radius of the circle: 1

Area of the circle is: 3.14

Circumference of the circle: 6.28

RESULT

Thus, the program in python to find the area and circumference of the circle was written and executed successfully.